Maxillary premolars

By

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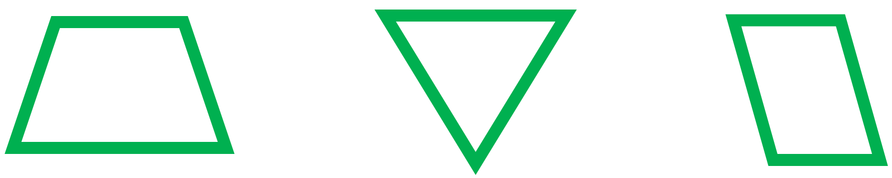
General Characteristics & Functions:

- 2 premolars per quadrant
- At least 2 cusps, buccal cusp always larger
- Distal to the canines
- Succedaneous
- Shorter crown than anterior teeth
- ‘Transitional tooth’ during chewing, or mastication
- It has properties of both the anterior canines and molars
- Assist molars in grinding, and canines in tearing
- Support facial muscles (aesthetics and speech)

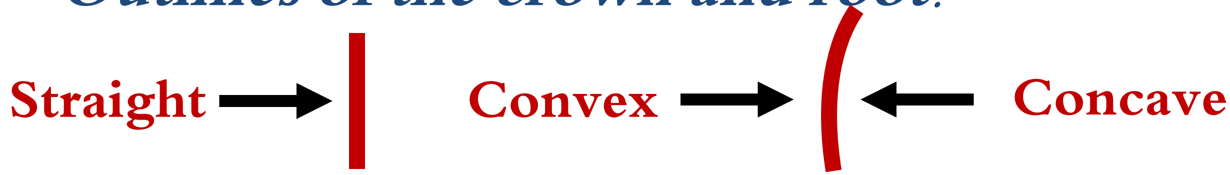
For easy teeth description

We have to speak about :

- *Geometric outline of the crown.*



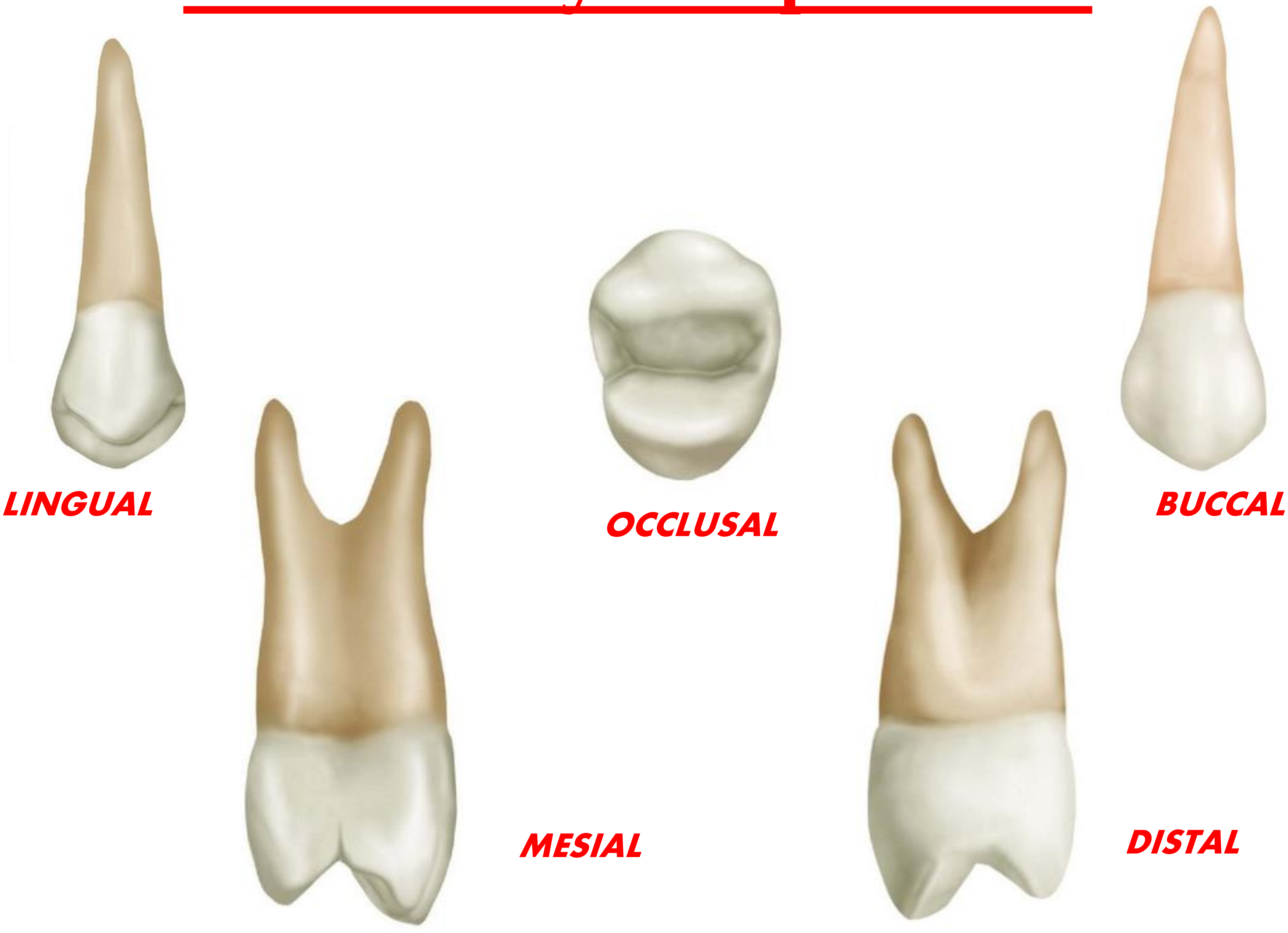
- *Outlines of the crown and root.*



- *Surface anatomy of the crown and root.*

(anatomical landmarks)

1- Maxillary first premolar



1- Maxillary first premolar

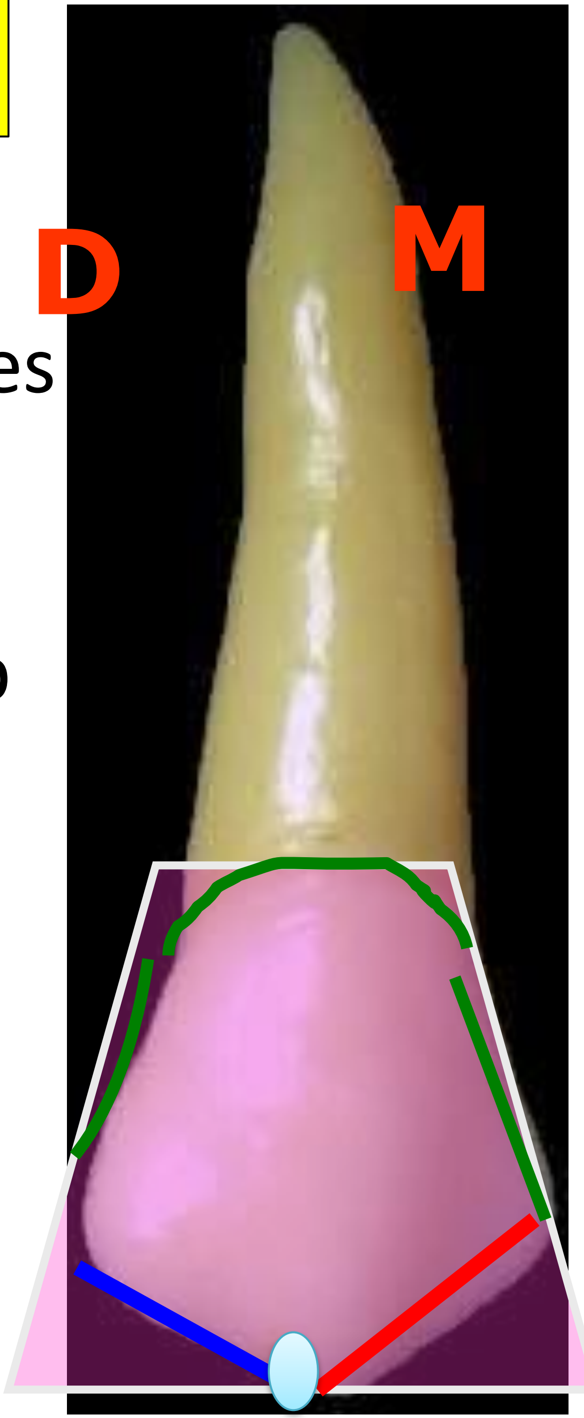
Principal identifying features

- Two roots, buccal and palatal (tend to curve distally)
- Two sharply defined cusps (buccal larger than palatal)
- Mesial developmental depression
- Mesial slope of buccal cusp longer than distal slope
- Palatal cusp tilts slightly mesially

• Buccal view features:

The geometrical out line : ***Trapezoid.***

- From buccal aspect, crown resembles maxillary canine ***except that*** mesial slope is longer than distal slope .
- Buccal cusp longer than palatal cusp
- The cusp tip is distally located.
- Mesial outline is slightly concave till the contact area.
- The distal outline is slightly concave till the contact area .
- Cervical line is convex rootwise.



Surface anatomy

- The buccal surface is convex.

The elevations

- The cervical ridge.
- The buccal ridge: well-developed middle lobe.

The depressions

Two developmental grooves mesial and distal to the buccal ridge.

The root

Buccal root is longer than palatal (if 2 roots)

The root gradually taper to the apex which is pointed and curved distally.

• Lingual view features

- Lingual surface is rounded in all directions and smaller than buccal surface (lingual convergence)
- The lingual cusp is shorter than the buccal cusp by 1mm.
- Mesial cusp slope of lingual cusp is shorter than distal cusp slope.
- Lingual cusp offset towards mesial.
- The lingual root is shorter and narrower than buccal root.

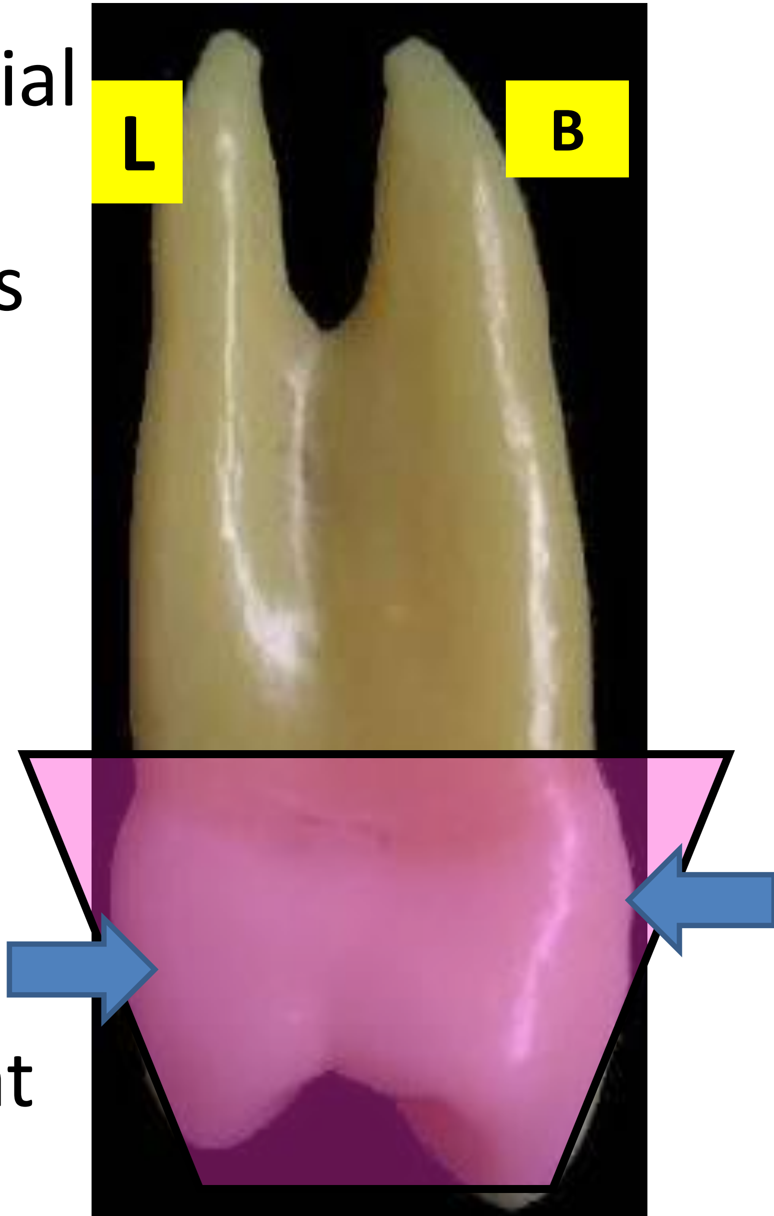
• Mesial aspect

✓ **Geometric outline** of the mesial aspect Trapezoid in shape Smallest of the uneven sides is occlusaly.

✓ **The outline**

✓ The buccal outline is convex till the buccal cusp tip with the maximum convexity at cervical 3rd.

✓ The lingual outline is convex with the maximum convexity at the center.



- ✓ The lingual cusp is shorter than the buccal cusp by 1 mm.
- The mesial contact area present at the middle third.

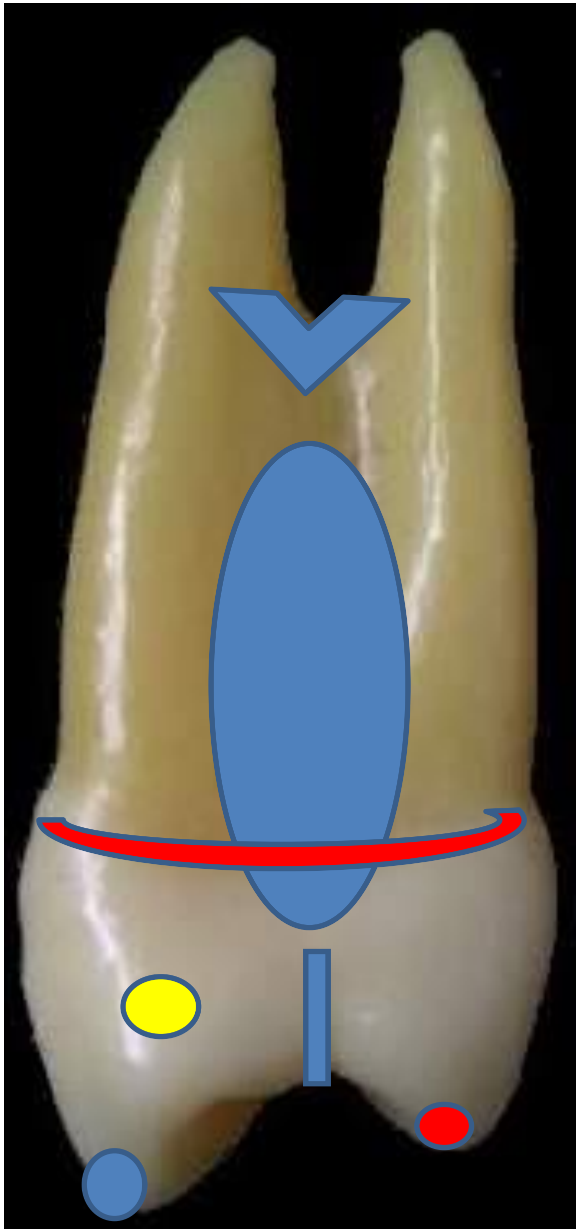
Surface anatomy

1- Mesial developmental groove crossing the mesial marginal ridge.

2-Mesial developmental depression above the contact area and continue on the root surface (The canine fossa).

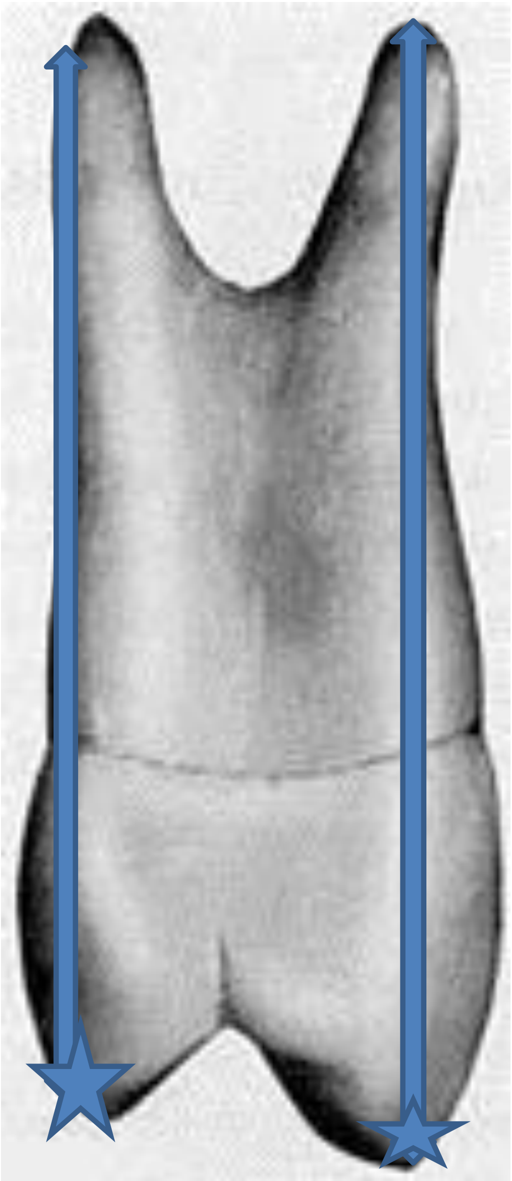
The root

- ✓ The bifurcation of the root present near the middle of the root length.



• Mesial aspect

- ✓ The buccal cusp tip is in one line with the center of the buccal root.
- ✓ The lingual cusp tip is in one line with the lingual border of the lingual root.



• Distal aspect

✓ **The distal surface differs than the mesial in:**

1- No developmental groove crossing the distal marginal ridge.

2- No developmental depression on the crown or root.

3- The contact area is broader.

4- Curvature of cervical line is less as compare to mesial surface it is almost flat.

5- The bifurcation present at apical third.

NOTE:

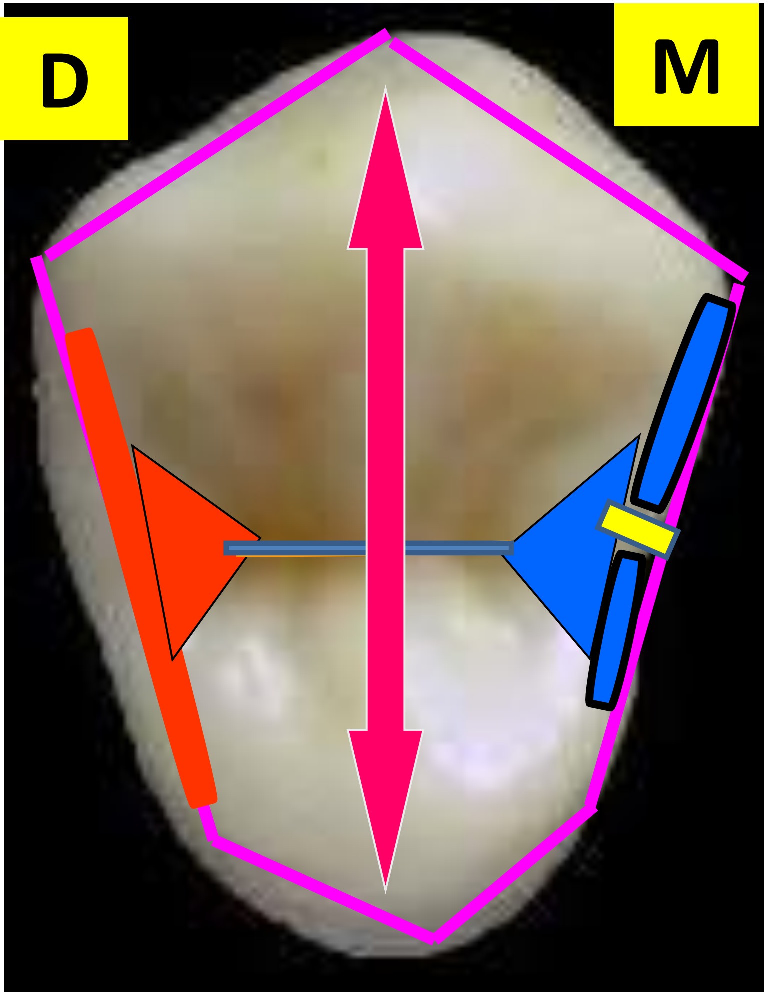
- ✓ Due to its long buccal root with narrow root canal and short palatal root with wide root canal, the upper 1st premolar is very prone to fracture during exodontia, hence, it is sometimes referred to some dentists as the ***"King of Fracture"***.

• Occlusal aspect

✓Geometric outline of the occlusal aspect is hexagonal in shape .

✓The crown is wider buccally than lingually.

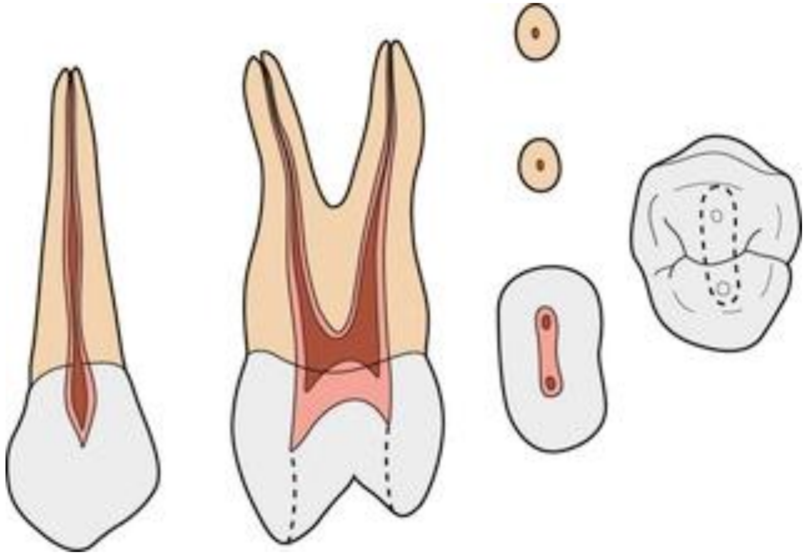
- 2 equal buccal sides (MB, DB)
- M side shorter than D side.
- ML side shorter than DL side.
- Thickness is greater than width
- CDG divides the occ. surf. evenly.
- MMR & DMR .
- MMR is crossed by MMRDG.
- Mesial and Distal triangular fossae are present .



Pulp cavity

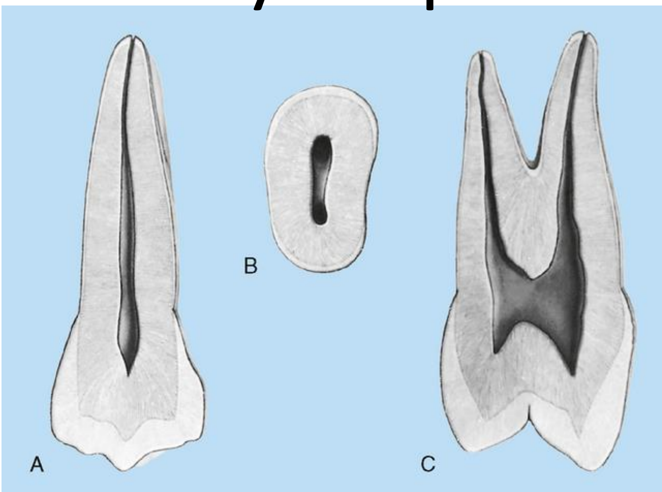
Bucco-lingual section

- ✓ Has wide pulp chamber with two pulp horns.
- ✓ There are two root canals tapering to the apical foramen
- ✓ In case of one root ; it has two root canals.
- ✓ The palatal root canal is wide.



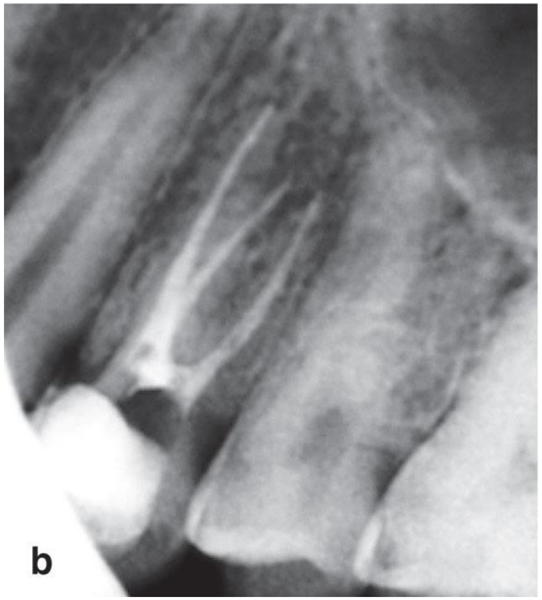
Cervical section:

- ✓ Kidney shape.



Mesio-distal section

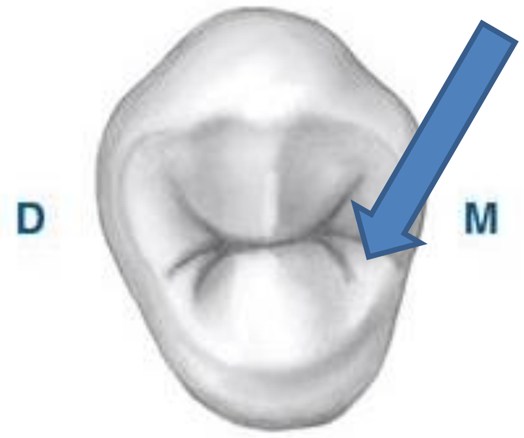
- ✓ The pulp chamber is bulbous.
- ✓ The root canal is narrow to the apex.


b


a

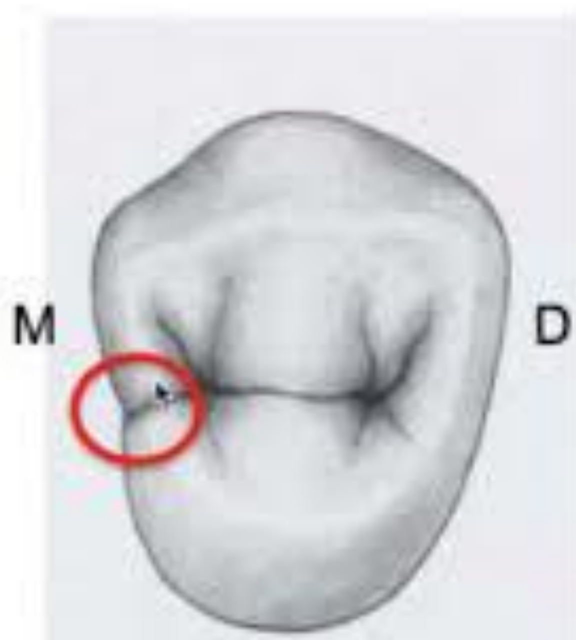

Q&A

- ✓ **The blue arrow is pointed to ----------------------**



- ✓ **The goemetrical out line of this aspect is ------------------------**

- ✓ **The red circles are showing------------------- ---------------------**
- ✓ **Which is the characteristic feature of ------------------ -----------------------**



- **Canine fossa is found on :**

A) **The mesial aspect of upper canine.**

B) **The mesial aspect of upper 5.**

C) **The mesial of upper 4.**

D) **The distal aspect of upper canine.**

Molarization

- Sometimes premolars have a root morphology similar to that of molars, a phenomenon known as molarization. Thus lower premolars will have a mesial and a distal root just like lower molars, and upper premolars have two buccal roots and one palatal root just like upper molars.

- The crowns in these premolars with molarization usually look quite normal, particularly in the upper premolars. Sometimes there may be an extra cusp present and the crown may be slightly longer mesio-distally.

- The frequency of molarization in premolars is approximately 1%. In the maxilla it is more frequent in the first premolar whereas in the mandible it is more frequent in the second premolar.


Thank you!!! ...